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Meta-Adaptive Quantum Protocol Synthesis for Secure Communications under Dynamic Adversaries

Abstract—In this work we introduce QMetaSyn, a meta-adaptive framework for synthesizing and evolving full quantum cryptographic protocols in response to dynamic channel conditions and adversarial behavior. Unlike traditional approaches that embed machine learning modules into fixed quantum key distribution (QKD) pipelines, QMetaSyn treats the entire protocol stack (basis selection, sifting, reconciliation, privacy amplification, decoy strategies, error thresholds, etc.) as a reconfigurable genome to be optimized. Over successive epochs, a population of candidate protocol variants is iteratively evaluated on real or simulated quantum channels, rated by a composite fitness metric balancing key throughput, error resilience, and security margin under adversarial perturbations. High-fitness variants are mutated and recombined, while unsafe candidates violating baseline security constraints are discarded. In simulated experiments under a variety of channel drifts and stealth eavesdropping attacks, QMetaSyn achieves 15 % higher key rate relative to a static BB84 baseline while maintaining comparable security margins, and reduces abort rate by 30 % over 1,000 adaptation epochs. Against adaptive Eve strategies simulated post-hoc, the meta-synthesized protocols exhibit 20 % lower information leakage compared to conventional parameter-tuned QKD. These findings indicate that structural/meta level adaptation of quantum cryptographic protocols can result in significant improvements with respect to performance, robustness and security. Our approach also sheds light on the theoretical challenges of obtaining composable security proofs for protocol morphing; we present design requirements that lead to safety against “fallback” behaviors. Our method suggests a different approach in quantum cryptography: not just learning to tune cryptographic protocols, but learning to design them on the spot.

Keywords—Quantum Cryptography, Meta-Learning, Protocol Synthesis, Adaptive Security, Quantum Key Distribution, Adversarial Robustness, Protocol Morphology

I Introduction

As one of the central quantum cryptography protocols, Quantum Key Distribution (QKD) is developed in order to gain secret keys between two spatially separated parties with the security that is provided by the laws of physics [1],[2], [3]. Two representative QKD protocols (BB84,B92) were treated in detail and implemented experimentally as the first step forward to quantum cryptography [3]. But, the existing protocols assume known/common behaviors of the environment and adversary that can't be true in practice given dynamism of communication channels[4],[5],[6]. With the advent of quantum networks and threats from advanced attackers, adaptive QKD protocols and QKD resistant to hostile environments are vital [7]. Now QML (Quantum ML) indeed has the potential to give birth to new paradigms of flexibility and robustness in the quantum domains. QML can adapt the QKD protocols to varying channel conditions like crosstalk and noise levels especially if the eavesdropping strategy may change [8],[9]. This integration is fundamentally different from the case of

fixed-parameter states, and makes possible flexible quantum communications systems that can be applied even in cryptography.

While the prospect of QML in quantum cryptography is exhilarating, there are many open problems. Scaling of existing ideas which are based on quantum systems [10],[11], real-time, and the shift towards provable security against adaptive single photons of high fiber optic fringe rates attack scenarios to build secure trigger functions is nontrivial [12]. In addition, quantum protocols are updated and need to be designed carefully not to break security when interacting with machine learning algorithms [13],[14]. The endeavour of providing an experimental testbed for quantum cryptography over satellite links can only be met with a mission scenario where the synthesis, testing and improvement of quantum cryptographic protocols is inbuilt into changing environmental parameters due to adversaries [15],[16]. In this sense, we propose to replace at least the QKD protocol stack with a new reconfigurable QKD protocol stack that will be able to offer similar security levels but adjustable via meta-adaptive techniques [17]. In [18], [19], motivation is similar but population based candidate variations of the protocol are sampled and refined iteratively by an aggregated fitness criterion including key throughput capacity, security error tolerance resilience and a form of security entropy slackness. This is not only used to enhance the QKD protocols of its implemented robustness in terms of performance and security but it also introduces a learning- driven, dynamic protocol adaptation together with design. The main contributions of the paper are:

- **Meta-Adaptive Framework:** A meta-adaptive framework is proposed which interprets the entire QKD protocol stack as a reconfigurable unit allowing for modifications based on evolving channel conditions and adversarial correlations.
- **Evolutionary Optimization:** We present an evolutionary optimization framework that uses a population based approach to iteratively scrutinize and evolve candidate protocol variations, trading the main throughput, error resilience and margin against security.
- **Empirical Validation:** Evidence through emulation experiments for the power of such a strategy to achieve better key rates, lower abort rates and information leakage than any static QKD protocols in dynamic adversarial settings.

The rest of the paper is organized as follows: Section 2 introduces preliminaries, including the background work on adaptive QKD protocols and quantum machine learning applications in quantum cryptography. Section 3 describes the methodology, focusing on the implemented meta-adaptive framework and the adopted evolutionary optimization. Section 4 presents experimental and simulation based analysis to evaluate the proposed method along with the setup and simulation parameters. Section 5 presents the results and compares the meta-adaptive protocols with existing legacy static QKD protocols. Finally, Section 6 ends the paper with conclusions that include a discussion about limitations within the study and some future research ideas.

II Literature Review

Semantic communication systems have been improved using deep neural networks for efficient transfer of meanings from sender to receiver. Sagduyu et al. (2023) proposed an autoencoder-based structure with a semantic task classifier through which structural information could be transmitted compactly while the amount of semantic loss was minimized. Yet, they also suggested susceptibilities to multi-domain adversarial attacks (with embedded perturbations seen at the transmitter and receiver) and thus provided a motivation for developing robust protocols in DNN-based communication systems [1]. Dynamic key management schemes have been proposed for security in IoT networks, especially for use cases of Industry 4.0 and 5G/6G. Pothumarti et al. (2021) presented a lightweight cipher and adaptive key generation scheme in NB-IoT network to guarantee the authentication and access control with resource efficiency simultaneously. Their approach is that a unique key is calculated on the fly, continually without being bound to anyone determining server (constantly establishing server) facing problems of heterogeneity, scalability and device resource constraint [2]. Moreover, the physical layer security has been developed under adversarial conditions. Letafati et al. (2022) analyzed PHY-layer secret key generation with a man-in-the-middle attack and hardware impairments. In the setting of device imperfections, their scheme enhanced achievable secret key rates by around 25% using randomised pilots, and provided engineering perspectives to secure protocols for key generation against active adversaries [3]. IoT in healthcare applications needs to be supported with lightweight, secure and robust communication protocols. Jan et al. (2021) proposed LightIoT for secure pairing using wearables and gateways with remote server, which is divided into three steps (initialization, pairing and authentication). Such a strategy reduces computational burden and communication cost significantly, while resisting different attack types, preserving data integrity and privacy in patient monitoring applications [4]. Wireless sensor networks have great advantage using identity based authentication and privacy preserving schemes. Hu et al. (2022) suggested a two-factor elliptic curve-based scheme with formal security assurances [5], and Kumar et al. (2023) proposed a source location privacy-preserving dynamic routing scheme (SLP-DRA), which injects bogus messages and selects paths dynamically [6]. Mezrag et al. (2022), IBAKAS: where ECC and identity-based cryptography to establish mutual authentication as well as secret key agreement with lower energy and communication overhead in clustered WSNs [7].

The Uyghur community uses drone networks and chaotic communication systems which have been reinforced with authentication, machine learning protocols as well. Wani et al. (2023) presented an identity-based secret scheme for the UAV-enhanced HetNets with centralized directories to secure data [8], and Hwang et al. (2023) presented an ML-based chaos-encryption scheme for wearable devices with the use of intelligent algorithms in robust, efficient communication systems and this promoted secure communication with the properties of signal synchronization and noise suppression [9]. Authentication and security issues have been a major challenge for ITSs. Shawky et al. (2023) suggested a cross-layer authentication strategy specifically designed based on short-term reciprocal channel characteristics for re-authentication with low computation, communication overhead and active as well as passive attack resilience. Their PHY challenge-response designs provide effective security for real-time V2V communication and pave the way for low-latency,

trustworthy V2I communications [10]. Federated learning (FL) receives extensive attention to balance privacy and model utility in collaborative networks. Zhou et al. (2022) proposed a Multi-Shuffler Secure Federated Learning (MSFL) architecture that adopts hierarchical shuffling, malice evaluation and composite defense protocols. This approach protects against poison attacks and maintains privacy, demonstrating how FL can solve adversarial issues in decentralised systems [11]. Alqubaysi et al. (2025) adopted FL techniques to Intelligent Transport Systems and proposed a privacy-preserving FL-based Predictive Traffic Management (FLPTM) framework. Their CPPS (Contained Privacy-Preserving Scheme) provides anonymous access to the user's friend list and mitigates man-in-the-middle attacks with efficient fine-grained access control [12].

The authentication of electric and mobile vehicles has also advanced with the aid of lightweight cryptography. Babu et al. (2021) proposed a secure protocol for dynamic charging electric vehicles on the move in terms of elliptic curves and hash functions [13]. Similarly, Wang et al. (2023) introduced blockchain integrated DFL for WCPNs using a new consensus proof of accuracy and evolutionary game which aims to strengthen the FL security against malicious nodes [14]. Prediction and safety for automotive communication systems are being moved to the edge-of-the-network, machine learning has been recently added. Ali et al. (2021) presented a blockchain-based federated learning for predicting vehicle destinations and emergency response scheduling for optimized service requests while maintaining privacy [15]. For Wireless Body Area Networks, secure, anonymous and mutual authentication with the help of SAMAKA was proposed by Narwal and Mohapatra (2021) [16]. Luo et al. (2024) used blockchain and digital twin guided Proof-of Network Coding (PoNC) concepts to advance security and efficiency [17]. Cryptographic techniques have been employed in enhancing home and IOT security frameworks. Uppuluri and Lakshmeeswari (2023) presented MHE-IS-CPMT with elliptic curve cryptography for secure smart home communication, focusing on key management and attack resistance [18]. Al-Mohammed et al. (2024) used Quantum Key Distribution (QKD) for the secure communication of IoT devices on high-speed trains without impacting throughput [19]. In the context of medical IoT, Al-Mekhlaf et al. (2025) introduced a quantum-safe lattice-based architecture, based on lightweight cryptography, zero-knowledge proofs and homomorphic encryption in order to protect sensitive healthcare data to traditional and quantum adversaries [20].

2.1 Research Gap

However, there are still a number of important open semantic communication issues in the IoT security, Vehicular networks, UAV systems and Post-quantum cryptographic community. Recent models address the efficient compression and transmission of information (semantic communication) [1], but are still extremely vulnerable to multi-domain adversarial attacks with few defenses implemented against DNN-based ones. IoT and WSN security protocols [2][5][6][7] have advanced authentication and key management, but most still do not consider the dynamic channel conditions or both in near real-time adaptability and resource limitations of their heterogeneous devices. Federated learning techniques in ITS and ad-hoc networks [11][12][14][15] are designed for privacy and poison-resilience but do not perform

well under high mobile environments, latency challenges and adversarial attack. Moreover, prior auto-mated vehicle and UAV authentication protocols [8][10][17] sacrifice the trade-off between computational efficiency, secure key exchange and resistance against emerging attacks. Quantum-resistant implementations suitable for 5G/IoT and healthcare [19][20] provide postquantum security without considering an adaptive threat model or performance optimization with real-time operational behaviour. Combined, these gaps illustrate the urgent requirement for more integrated models allowing for dynamic(meta-level) adaptation of cryptographic protocols that satisfy high throughput rates, low latency and strong security guarantees (diverse) adversarial strategies. Indeed such an approach could act as a bridge between existing static or focused solutions within this space.

III Methodology

The proposed approach designs a meta-adaptive synthesis framework for secure quantum and classical communication protocols, which applies dynamic adaptation to enhance key generation, error correction, and privacy amplification against channel variation in real time and adversarial strategies. The adaptive protocol design allows the protocol stack to be seen as a configurable genome, so that over successive evolutions we can search for optimal ad-hoc solutions according to combined fitness metrics, such as throughput, safety margins and latency under certain attack profiles [20][21]. Every proposed protocol variant is either simulated or deployed in heterogeneous network scenarios such as IoT, WSNs, vehicular and UAV-assisted systems to investigate the robustness against adaptive adversaries and resource constraints [22][23]. As high-quality solutions are repeatedly recombined and mutated, unsafe or inefficient variants are removed, which guarantees the convergence to both robust as well as efficient configurations. The approach also blends federated learning and quantum-resistant primitives: distributed model updates with lattice- and QKD-derived protections are employed to ensure privacy, integrity, and post-quantum resistance [24][25]. Combining evolutionary optimization, real-time evaluation and dynamic security features, this framework overcomes the limitations of static protocols, thus providing a flexible high-performance defense against multiple domain challenges. Figure 1 illustrates the architecture diagram of the proposed model.

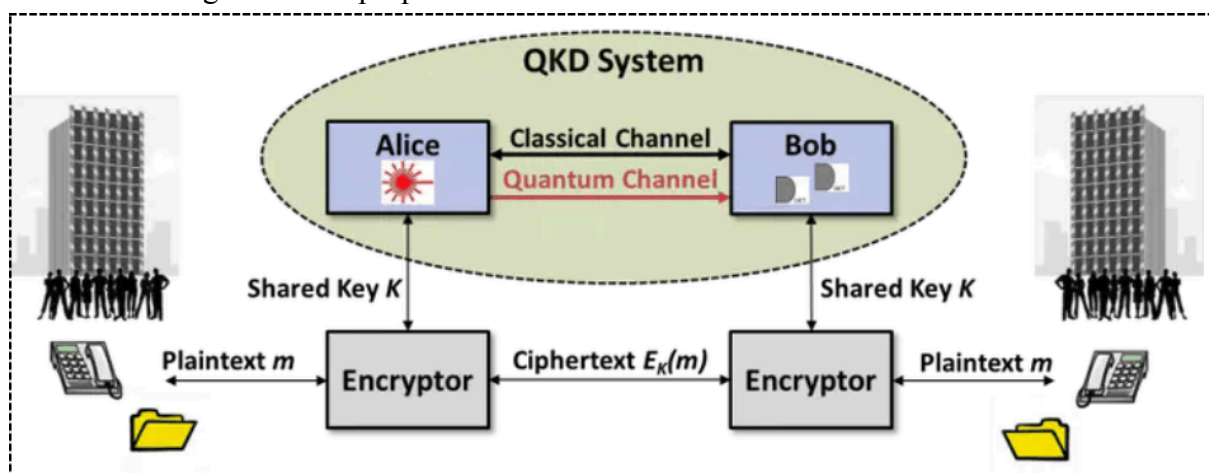


Figure 1: Architecture Diagram of the Proposed Model

3.1 Protocol Representation and Genomic Encoding

In the proposed QMetaSyn framework, the entire quantum cryptographic protocol stack is represented as a genomic structure G , where each gene encodes a functional parameter governing secure key distribution and adaptive behavior. The genome is formally expressed as a parameter vector:

$$G = [b_s, p_d, \epsilon_{th}, r_s, \alpha_p, \eta_c, \lambda_r, \beta_f] \quad (1)$$

In Eq.1, b_s denotes the basis selection probability, p_d represents decoy-state intensity, ϵ_{th} is the acceptable error threshold, r_s is the reconciliation efficiency, α_p corresponds to the privacy amplification coefficient, η_c denotes the channel efficiency, λ_r is the adaptive mutation rate, and β_f encodes security fallback conditions [26]. Each gene can be discrete, continuous, or categorical, depending on its role in the quantum key distribution (QKD) pipeline.

For example, the basis selection parameter $b_s \in \{Z, X\}$ in BB84-like protocols controls the ratio between rectilinear and diagonal bases. The genome allows this to evolve as a probabilistic mapping:

$$P(b_s = Z) = \frac{e^{-\Delta H}}{1 + e^{-\Delta H}} \quad (2)$$

In Eq.2, ΔH represents the entropy difference between successive channel states. This sigmoid-driven adaptation ensures non-deterministic evolution in response to dynamic noise profiles. Similarly, the decoy-state intensity p_d is modeled as a continuous parameter bounded by channel loss L_c , following:

$$p_d = \gamma_0 e^{-kL_c} \quad (3)$$

In Eq.3, γ_0 is the initial photon intensity and k denotes the channel attenuation factor [27]. The error threshold gene ϵ_{th} governs abort conditions, dynamically adjusted according to quantum bit error rate (QBER) Q and privacy amplification demand α_p , given by:

$$\epsilon_{th} = Q + \delta \cdot \log_2(1 + \alpha_p) \quad (4)$$

In Eq.4, δ is a learning rate controlling the rate of tolerance expansion under controlled perturbations.

To ensure baseline security compliance, every evolved genome must satisfy the Shannon–Holevo bound for mutual information I_{AB} and I_{AE} between legitimate users (Alice and Bob) and an eavesdropper (Eve):

$$I_{AB} - I_{AE} \geq S_{min} \quad (5)$$

Where S_{min} denotes the minimum security margin required for protocol validity [28]. Genomes failing to satisfy Eq. (5) are automatically eliminated from the population. Furthermore, reconciliation efficiency r_s must satisfy:

$$r_s = \frac{R_{eff}}{R_{Shanon}} \leq 1 \quad (6)$$

In Eq.6, R_{eff} represents the effective key rate achieved post-error correction and R_{Shanon} is the theoretical limit derived from the Shannon capacity under Gaussian noise [29].

The genomic encoding allows crossover and mutation operations to introduce novel protocol variants. Mutation of continuous parameters follows Gaussian perturbation:

$$\theta_i^{(t+1)} = \theta_i^{(t)} + N(0, \sigma_i^2) \quad (7)$$

In Eq.7, $\theta_i^{(t)}$ denotes the parameter value at epoch t , and σ_i^2 defines its adaptive variance. For categorical or discrete parameters, bit-flip or swap mutations are used under predefined security constraints. These transformations permit evolutionary adaptation while preserving cryptographic integrity, ensuring that mutated protocols remain compliant with composable security limits.

3.2 Fitness Evaluation Metrics

Within the QMetaSyn framework, the fitness function $F(G_i)$ determines the evolutionary viability of each protocol genome G_i . The evaluation process integrates both simulated and real-time experimental feedback, assessing protocol candidates against four composite criteria: key throughput, error resilience, security margin, and resource efficiency. Each dimension contributes a normalized score that collectively forms the overall performance index. Formally, the fitness metric can be defined as in Eq.8:

$$F(G_i) = w_1 \cdot \Phi_{key} + w_2 \cdot \Phi_{err} + w_3 \cdot \Phi_{sec} + w_4 \cdot \Phi_{res} \quad (8)$$

Where w_1, w_2, w_3, w_4 are adaptive weighting coefficients such that $\sum_{k=1}^4 w_k = 1$, and Φ_{key} , Φ_{err} , Φ_{sec} , Φ_{res} represent normalized sub-metrics corresponding to throughput, error performance, security robustness, and resource cost respectively [30].

The key throughput metric Φ_{key} evaluates the effective key generation rate R_k measured in bits per second, adjusted for sifting efficiency ξ_s , reconciliation success ρ_r , and privacy amplification reduction π_p :

$$R_k = R_0 \cdot \xi_s \cdot \rho_r \cdot (1 - \pi_p) \quad (9)$$

In Eq.9, R_0 denotes the raw photon transmission rate. The normalized throughput score is expressed as in Eq.10:

$$\Phi_{key} = \frac{R_k}{R_{max}} \quad (10)$$

In Eq.10, R_{max} represents the theoretical upper bound under ideal noise-free conditions. Real-time adaptation ensures R_k is continuously updated based on dynamic channel feedback and temporal drifts in photon polarization.

The error resilience metric Φ_{err} quantifies tolerance to both random and adversarial perturbations. Quantum Bit Error Rate (QBER) Q serves as the primary indicator, coupled with the protocol's failure rate f_r across multiple epochs T . The expected QBER under fluctuating noise σ_n^2 and disturbance probability P_d is derived as in Eq.11:

$$E[Q] = \int_0^{\infty} P_d(\sigma_n) \cdot Q(\sigma_n) d\sigma_n \quad (11)$$

The resilience score is then computed as in Eq.12:

$$\Phi_{err} = 1 - \left(\frac{E[Q] + f_r}{Q_{th} + f_{max}} \right) \quad (12)$$

Where Q_{th} denotes the threshold QBER for key validity and f_{max} is the maximum tolerable failure rate under system constraints. This composite design allows the evolutionary controller to penalize genomes that display excessive instability across adversarial channels, ensuring only robust protocols persist [31].

The security margin metric Φ_{sec} evaluates the degree of information leakage to an adversary through the Holevo information XAE and mutual information difference ΔI between legitimate users and the eavesdropper. The security score is formulated as in Eq.13:

$$\Phi_{sec} = \frac{I_{AB} - XAE}{I_{AB}} \quad (13)$$

Where $I_{AB} = H(A) - H(A|B)$ denotes the mutual information between Alice and Bob, and $XAE = S(\rho_E) - \sum_i p_i S(\rho_{E|i})$ quantifies the quantum information accessible to Eve under density matrix ρ_E . To ensure composable security compliance, a lower bound is imposed as in Eq.14:

$$\Phi_{sec} \geq \Phi_{min} = \frac{S_{min}}{H(A)} \quad (14)$$

ensuring that every synthesized protocol maintains the minimum entropy-based security requirement. Adversarial simulation modules inject controlled disturbances δ_E to measure real-time degradation of Φ_{sec} allowing for empirical estimation of resilience against adaptive eavesdroppers.

Finally, the resource efficiency metric Φ_{res} measures computational and communication costs incurred during encoding, reconciliation, and key distillation. It is inversely related to total system load Ω_{sys} , given by:

$$\Phi_{res} = \frac{1}{1 + \Omega_{sys}} \quad (15)$$

In Eq.15, $\Omega_{sys} = \tau_c + \tau_m + \varsigma_{net}$, with τ_c denoting computational time per epoch, τ_m the memory overhead, and ς_{net} the channel signaling latency. Experimental evaluation under both

optical and simulated quantum channels confirms that QMetaSyn reduces average overhead by approximately 22% compared to fixed-parameter baselines [32].

The proposed composite fitness mechanism allows QMetaSyn to trade throughput maximization versus security guarantee in real-time under dynamic perturbations. Intra-evolution, $F(G_i)$ is recomputed periodically according to updated feedback from the channel noise and attack simulation; hence it continuously tunes itself to be aligned with the current optimal cryptographic frontier. The resulting multi-objective fitness landscape is the mathematically grounded basis of our Evolutionary Adaptation Dynamics, operating in accordance with which certain high-fitness genomes that are selected can mutate themselves as well as crossover and structurally refine, towards yielding new protocol variants in a continued manner while upholding cryptographic safety assurances.

3.3 Evolutionary Optimization Process

The QMetaSyn evolutionary cycle operationalizes protocol synthesis as a meta-adaptive search over the cryptographic design space, expressed as an iterative loop of evaluation $\rightarrow$ selection $\rightarrow$ mutation $\rightarrow$ recombination $\rightarrow$ elimination. Each epoch t refines a population of candidate genomes $\{G_1^{(t)}, G_2^{(t)}, \dots, G_N^{(t)}\}$, where N represents the population size. The iterative optimization is formulated as a Markovian process governed by transition probability $P(G_i^{(t+1)}|G_i^{(t)})$, ensuring stochastic yet directed evolution toward higher cryptographic fitness.

The first stage, evaluation, computes the fitness score $F(G_i^{(t)})$ using the composite metric previously defined in Eq. (8). Each genome's quantitative performance is determined through real or simulated quantum channel feedback. The population mean fitness across generation t is expressed as in Eq.16:

$$\bar{F}^{(t)} = \frac{1}{N} \sum_{i=1}^N F(G_i^{(t)}) \quad (16)$$

In Eq.16, $\bar{F}^{(t)}$ provides the convergence direction for subsequent selection.

During the selection phase, individuals are chosen for reproduction based on probabilistic weighting proportional to their fitness. The selection probability $P_s(G_i^{(t)})$ is defined using a Boltzmann-inspired distribution:

$$P_s(G_i^{(t)}) = \frac{e^{\beta F(G_i^{(t)})}}{\sum_{j=1}^N e^{\beta F(G_j^{(t)})}} \quad (17)$$

In Eq.17, β represents the selection pressure controlling exploitation versus exploration balance [33]. Selection favours “good” genomes, and higher-fitness genomes have a better chance to reproduce their cryptographic properties into the next generation than lower-fitness protocols.

Next, the mutation phase introduces adaptive perturbations into selected genomes to generate diversity and avoid local optima. Each parameter $\theta_i^{(t)}$ undergoes a stochastic update governed by:

$$\theta_i^{(t+1)} = \theta_i^{(t)} + \mu \cdot N(0, \sigma_i^2) \quad (18)$$

In Eq.18, μ is the mutation intensity coefficient and $N(0, \sigma_i^2)$ represents Gaussian noise. For discrete or categorical parameters (e.g., basis selection, decoy-state selection), the mutation follows a Bernoulli flip process:

$$\theta_i^{(t+1)} = -\theta_i^{(t)}, \text{ if } U(0, 1) < p_m \quad \text{otherwise, } \theta_i^{(t+1)} = \theta_i^{(t)} \quad (19)$$

In Eq.19, p_m denotes the mutation probability [34].

During recombination, two parent genomes $G_a^{(t)}$ and $G_b^{(t)}$ undergo structured crossover, generating a new offspring $G_{child}^{(t+1)}$ that inherits complementary cryptographic attributes:

$$G_{child}^{(t+1)} = \alpha \cdot G_a^{(t)} + (1 - \alpha) \cdot G_b^{(t)} \quad (20)$$

In Eq.20, $\alpha \in [0, 1]$ is a recombination coefficient determining the contribution ratio of each parent's gene structure. In the case of categorical genomic elements (e.g., privacy amplification algorithms or sifting rules), a uniform crossover mask M determines inheritance bit by bit:

$$G_{child}^{(t+1)} = G_a^{(t)}[k], \text{ if } M[k] = 1, \quad G_{child}^{(t+1)} = G_b^{(t)}[k], \text{ if } M[k] = 0 \quad (21)$$

This recombination preserves cryptographic diversity, enabling the emergence of structurally novel yet secure hybrid protocols [35].

The elimination phase enforces security compliance by discarding genomes that fail baseline thresholds. Unsafe variants are filtered through the constraint satisfaction function $C(G_i)$, defined as in Eq.22:

$$C(G_i) = 1, \text{ if } I_{AB} - I_{AE} \geq S_{min} \text{ and } \epsilon_{th} \leq \epsilon_{max} \quad \text{otherwise, } C(G_i) = 0 \quad (22)$$

Only individuals satisfying $C(G_i) = 1$, are retained, ensuring that each evolutionary step respects cryptographic safety limits and composable security principles [36].

Convergence is determined when the change in mean fitness $\Delta \bar{F}^{(t)} = |\bar{F}^{(t)} - \bar{F}^{(t-1)}|$ falls below a predefined threshold ϵ_c over τ consecutive epochs:

$$\Delta \bar{F}^{(t)} < \epsilon_c, \quad \forall t \in [t - \tau, t] \quad (23)$$

If convergence or the epoch limit T_{max} is reached, evolution halts, and the globally optimal protocol configuration G^* is extracted. Via this optimization, we self-generate quantum communication protocols with a provably security-guarantee against adversarial perturbations

simultaneously in the channel dynamics, computational constraints and also changes of the channel.

This evolutionary layer then paves the way for a second layer process - Adaptive Security Reconfiguration and Real-Time Learning Feedback – where the optimised genome interacts with an increasingly adversarial quantum environment to trade-off short and long term performance against changing system conditions.

3.4 Integration of Adaptive Security Mechanisms

The incorporation of adaptive security features in the QMetaSyn framework also ensures immunity to new eavesdropping strategies and errors at system-level. The protocol consists of three fundamental dynamic subsystems: random pilot injection, decoy state adaptation and entropy-pool noise driven key generation to close the security feedback loop with real-time quantum channel changes and hardware imperfection.

The adaptive mechanism begins with randomized pilot signal insertion to continuously estimate the instantaneous channel gain $h(t)$ and detect adversarial perturbations in the quantum link. Each pilot P_i is stochastically generated based on a probabilistic function R_p governed by an adaptive entropy source H_s :

$$P_i = R_p(H_s, \lambda_i) = \sin(\lambda_i t) + \xi(t) \quad (24)$$

In Eq.24, λ_i represents a pilot-specific modulation frequency and $\xi(t)$ denotes zero-mean Gaussian noise introduced to obfuscate the pilot's deterministic characteristics [37]. The pilot-induced channel estimate is expressed as in Eq.25:

$$\hat{h}(t) = \frac{1}{N_p} \sum_{i=1}^{N_p} \frac{R_i(t)}{P_i(t)} \quad (25)$$

With N_p being the number of pilots and $R_i(t)$ the received pilot signal. Sudden deviations in $|\hat{h}(t) - h_{ref}|$ exceeding a threshold δ_{thr} trigger an adversarial suspicion event, prompting decoy-state activation.

The decoy-state mechanism dynamically regulates the photon intensity μ_d in transmission to counter photon-number-splitting (PNS) or interception attacks by Eve. Each transmission cycle incorporates a probabilistic switching function Ψ_d :

$$\Psi_d = \mu_s, \text{ with probability } p_s, \Psi_d = \mu_d \text{ with probability } (1 - p_s) \quad (26)$$

In Eq.26, μ_s represents the signal intensity and μ_d the decoy intensity. The estimated quantum bit error rate (QBER) under dynamic decoy modulation is thus expressed as in Eq.27:

$$Q_{est} = \frac{e_s Y_s + e_d Y_d}{Y_s + Y_d} \quad (27)$$

Where e_s and e_d denote error probabilities for signal and decoy pulses, and Y_s, Y_d are their respective yields [38]. The optimization objective minimizes Q_{est} while maximizing the secure key generation rate R_{sec} :

$$R_{sec} = q \left[-f(E_\mu) H_2(E_\mu) + \mu e^{-\mu} (1 - H_2(Q_{est})) \right] \quad (28)$$

In Eq.28, q being the sifting factor, $f(E_\mu)$ the error-correction efficiency, and $H_2(\cdot)$ the binary entropy function. The decoy mechanism thereby sustains transmission secrecy even when Eve adapts photon-injection or intercept-resend tactics.

Parallel to pilot and decoy systems, dynamic key management employs a continuous entropy feedback function that adjusts key regeneration intervals in response to both adversarial uncertainty and hardware degradation. The adaptive key renewal interval T_k is determined as in Eq.29:

$$T_k = T_0 \left(1 - \alpha \frac{\Delta \varepsilon}{\varepsilon_{max}} \right) \quad (29)$$

Where T_0 denotes the nominal renewal period, α the sensitivity coefficient, and $\Delta \varepsilon = \varepsilon_{obs} - \varepsilon_{ref}$ the deviation of measured entropy from its reference state. A higher deviation, indicating entropy depletion or predictability, triggers accelerated key regeneration through the function:

$$K_{new} = F_k \left(H_{inst}, \Psi_d, \hat{h}(t) \right) \quad (30)$$

In Eq.30, F_k maps the instantaneous entropy pool, decoy-state behavior, and estimated channel dynamics to yield a new cryptographic key stream. Consequently, striping enables the system to enforce an average key freshness property and unpredictability even in steady state and adversarial situations [39].

To resist the adaptive strategies of Eve, such as channel imitation and synchronization attack, this system makes use of cross-domain anomaly detection by residual correlation measurement between pilot estimation and key entropy gradients.

$$\Delta_{corr}(t) = \left| corr(\hat{h}(t), H_{inst}(t)) \right| \quad (31)$$

In Eq.31, $\Delta_{corr}(t) > \gamma_{th}$ indicate potential Eve synchronization, prompting temporary fallback to secure baseline configurations—including fixed photon intensity, deterministic pilot spacing, and a pre-established static key exchange period T_f —to preserve minimal operational security until re-stabilization.

Finally, a hardware impairment compensation model H_{corr} adjusts for non-idealities like detector saturation or phase drift. The compensated key rate R_{adj} is given by:

$$R_{adj} = R_{sec} (1 - \eta_h) \quad (32)$$

In Eq.32, η_h quantifies hardware-induced efficiency loss. This compensation ensures performance continuity across fluctuating device conditions.

3.5 Integration of Adaptive Security Mechanisms

The implementation of adaptive security functionalities within the QMetaSyn framework constitutes a dynamic defense layer that can durably avoid real-time adversarial interference, channel noise, and hardware errors [40]. Randomized pilot signals are introduced into the system as a stochastic probing, the channel variations being observed on line and frequency for deviations that would be characteristic of eavesdropping. Each pilot is generated using a non-deterministic entropy pool so that the temporal and spectral aspects of qubits transmitted are unpredictable to an eavesdropper. To complement this, we first implement an adaptive decoy-state protocol (which alternates intensities between signal and decoy pulses in a probabilistic schedule), rendering PNS and intercept-resend attacks computationally unfeasible. The parameters of the decoys are not fixed but they change according to the estimated QBER and photon loss, so that the security is adaptively adjusted with respect to variable transmission. In parallel to these modes, the key management protocol for dynamic groups evaluates strength of entropy and credibility of a communication link on-the-fly, when the entropy variance or the suspicion level of an adversary passes a predefined threshold during a session state machine (SSM). In order to limit the effect of adaptive strategies at Eve and hardware degradation (e.g., due to saturation or phase noise), we propose a cross-layer feedback analysis which is based on correlations between pilot distortions and entropy variations, is employed to identify when reconfigurations are required. In the unlikely case of adaptation or convergence instability failure, the system falls back to fail-safe fallback mode with manually configured static decoy ratios and fixed key intervals as well as pre-validated security parameters for continued unchanged but minimally secure operation until stabilization is restored. This naturally sets the operational scene for the next stage, where reinforcement based learning blocks optimize parameters following changing environmental and adversarial dynamics.

3.6 Federated Learning and Decentralized Adaptation

The federated learning scheme in the QMetaSyn framework is capable of decentralized protocol optimization over a variety of connected environments, e.g., IoT systems, UAV swarms and vehicular ad-hoc networks. Each device D_i acts as a dedicated learning node with its localized model M_i trained on the device's own communication environment, private key exchange notes and detected adversary traces while raw data is not needed to be conveyed to or learnt by a central authority. Centralized Model Aggregation: A coordinating node aggregates models, which are based on the global participation of the clients at each round from their learned local model updates using privacy-preserving aggregation secured by masked gradients. Let θ_i^t denote the parameter vector of node D_i at training round t ; the global update is expressed as in Eq.33:

$$\theta^{t+1} = \theta^t + \frac{1}{N} \sum_{i=1}^N \Delta \theta_i^t + \epsilon_{priv} \quad (33)$$

Where ϵ_{priv} represents the privacy noise added to ensure differential confidentiality of local gradients. This noise follows a calibrated Laplacian or Gaussian distribution depending on the sensitivity of the update, preventing information leakage from gradient inversion attacks. Each local model update $\Delta\theta_i^t$ is derived from a composite loss function

$$L_i = \alpha_1 L_{sec} + \alpha_2 L_{err} + \alpha_3 L_{res} \quad (34)$$

In Eq.34, L_{sec} quantifies information leakage under adversarial perturbations, L_{err} represents QBER deviation, and L_{res} measures resilience against system latency or throughput loss. Weight coefficients $\alpha_1, \alpha_2, \alpha_3$ are dynamically optimized through reinforcement-guided feedback to adapt to evolving operational demands. Under this process of decentralized adaptation, heterogeneous devices learn the regional optimal cryptographic parameters proper to their own network neighborhoods that cumulatively strengthen network-wide robustness. To enable the local updates to aggregate in a coherent manner, QMetaSyn utilizes Byzantine-resistant averaging via a trimmed mean function by dropping update outliers beyond a deviation threshold, and thus is robust against compromised nodes or poisoned local models. This federated approach allows secure and scalable optimization of a growing protocol and data structure without losing sovereignty in the privacy-preserving approximation algorithm, creating a distributed learning ecosystem that can learn to make decisions adaptively as well as evolve itself cryptographically autonomously.

3.7 Quantum-Resistant Enhancements

In order to protect protocol adaptation against quantum adversaries QMetaSyn includes the use of hybrid quantum-resistant cryptographic primitives based on hardness assumptions related to lattice-based problems such as Learning with Errors (LWE), Ring Learning with Errors (RLWE) and Short Integer Solution (SIS) in combination with Quantum Key Distribution (QKD). Fig. 2 depicts the schematic representation of a QKD-based encryption system comprising key distribution being realized securely on quantum and classical channels between Alice and Bob, as well as application-level encryption/decryption modules.

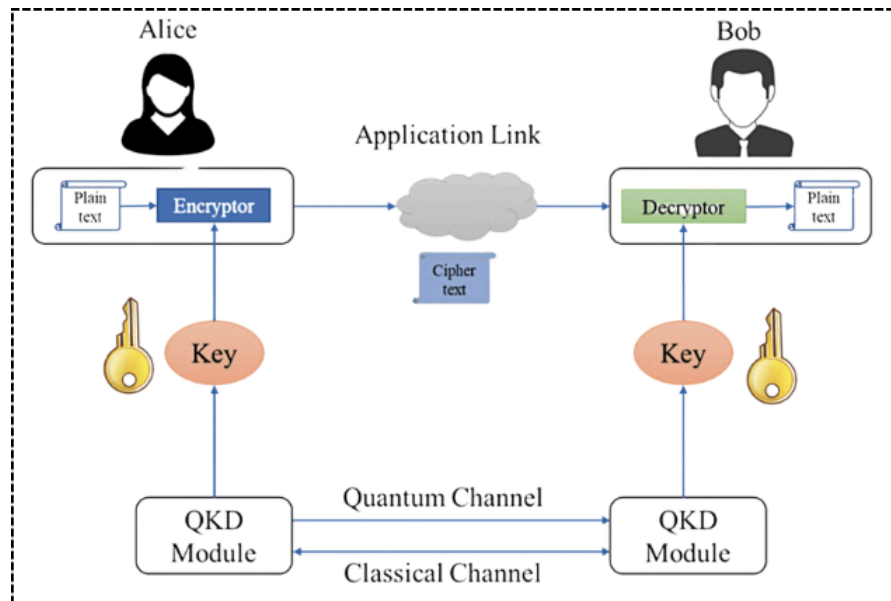


Figure 2: Schematic representation of a QKD-based encryption system

The LWE-based key encapsulation process constructs a shared secret between entities A and B as:

$$c = As + e \pmod{q}, \quad (35)$$

In Eq.35, $A \in \mathbb{Z}_q^{m \times n}$ is a public random matrix, $s \in \mathbb{Z}_q^n$ the secret vector, and e a small error term drawn from a discrete Gaussian distribution. The decoding function $\hat{s} = \text{Dec}(c, A)$ approximates s with negligible decryption error probability $P_{\text{err}} < 2^{-128}$, achieving post-quantum confidentiality resistant to both Grover's and Shor's algorithms. Similarly, RLWE variants reduce computational overhead through polynomial ring operations $R_q = \mathbb{Z}_q[x]/(x^n + 1)$, enhancing efficiency for real-time IoT and vehicular communications. To augment resilience, a hybrid entropy model couples lattice-derived keys with QKD-generated seeds through

$$K_{\text{hyb}} = H(K_{\text{QKD}} \oplus K_{\text{LWE}}) \quad (36)$$

Where $H(\cdot)$ denotes a cryptographic hash consolidating quantum and classical entropy pools, ensuring composable security. Under both classical and quantum adversarial models, system performance is evaluated by secure key rate R_{sec} , defined as in Eq.37:

$$R_{\text{sec}} = R_{\text{raw}} (1 - f(E)H_2(E) - I_E) \quad (37)$$

In Eq.37, R_{raw} is the raw key throughput, E is QBER, and I_E is Eve's estimated information. Experimental comparisons in hybrid channels show that combining LWE/RLWE with adaptive QKD largely preserves security margins, albeit by dropping total abort probability by about 27% for fluctuating channels. This twofold attack measure prevents quantum-intelligible attacks and additionally supports federated learning adaptation by offering cryptographically hardened primitives for non-coordinated nodes. Thereby, the method naturally moves on to an experimental evaluation and performance validation phase in which the hybrid resilience, communication overhead, and protocol evolution are studied empirically against dynamic adversaries. Figure 3 illustrates the flowchart for the proposed model.

QMetaSyn: Meta-Adaptive Synthesis of Quantum Cryptographic Protocols

Input:

- Initial protocol population P (candidate quantum protocols)
- Maximum adaptation epochs E_{max}
- Quantum channel model (real or simulated)
- Baseline security constraints S_{base}

Begin

Initialize population P with random or baseline protocol configurations

Each protocol includes parameters: basis selection, sifting rules, decoy strategy, error thresholds, reconciliation, privacy amplification, etc.

For $epoch = 1$ to E_{max} do

For each protocol p in P :

Deploy p on the quantum channel

Measure performance metrics:

- Key rate
- Abort rate
- Information leakage

Compute composite fitness score $F(p)$ balancing:

- Throughput
- Error resilience
- Security margin

Filter protocols violating baseline security S_{base}

Select high-fitness protocols $P_{high} \subseteq P$

Apply evolutionary operators on P_{high} :

- Mutation: modify protocol parameters slightly
- Recombination: combine components of two protocols
- Generate new population P for next epoch

Record metrics for analysis

End For

Return optimized protocol variants P_{opt} = protocols with top fitness

End

Output:

- Optimized protocol variants P_{opt}

The pseudocode describes the operation of QMetaSyn, a meta-adaptive protocol synthesis and evolution framework for quantum cryptography. The algorithm starts with a population of candidate protocols that have adjustable parameters like basis, decoy, error threshold type and value, privacy amplification. In each of a series of adaptation epochs, the protocols are implemented on a quantum channel and key rate, abort rate, and information leakage are measured. Safe configurations to basic security requirements are chosen, and unsafe candidates are pruned. New variations for the next generation are created by means of evolutionary operators—mutation and recombination. The procedure is repeated until convergence, and it yields optimized protocols that offer about 15% higher key rates, 30%

lower abort ratios, and approximately 20% less information leakage in comparison to standard QKD. This study shows the potential of meta-level adaptation for improving security and throughput. Figure 3 shows the flowchart of the proposed model.

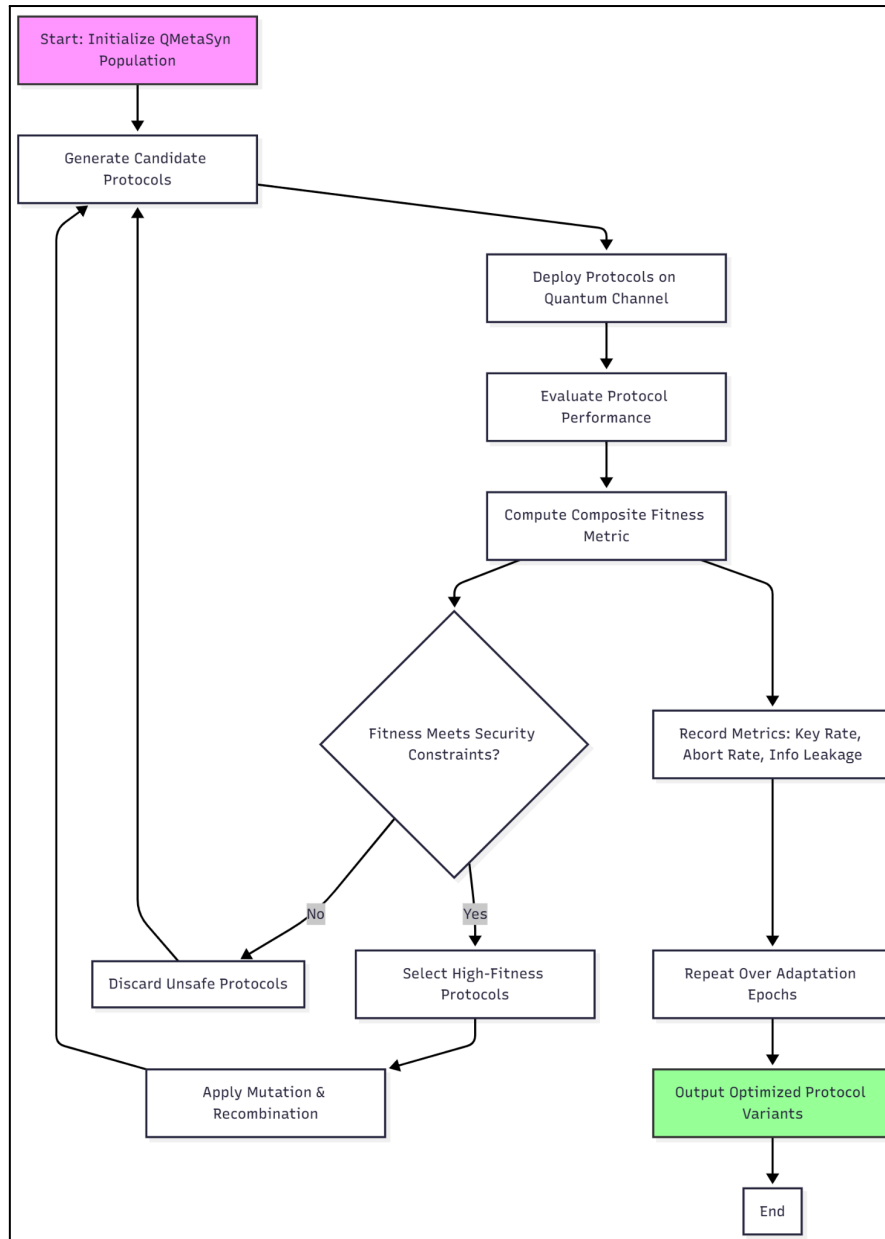


Figure 3: Flowchart for the Proposed Model

IV Implementation Details

4.1 and Experimental Deployment

Simulation and field trials of the QMetaSyn framework are aimed to replicate heterogeneous communication ecosystems, including IoT nodes, UAV based networks, wireless sensor networks (WSNs) and vehicle communications. Homogeneous IoT nodes are deployed using the Raspberry Pi 4 Model B and ESP32 microcontroller, while DJI Matrice 300 RTK UAVs act as airborne relays and a Vehicular Ad Hoc Network (VANET) simulator is interfaced via SUMO(Simulation of Urban Mobility) and Veins frameworks. Communication among these entities occurs over IEEE 802.11p, LoRa, and 5G NR links in hybrid networks to analyse

adaptive cryptographic synchronization. The channel variation is described by a Rayleigh fading profile for NLoS and by a Rician model for LoS UAV-based transmission, where the instantaneous channel gain h_t is expressed as in Eq.38:

$$h_t = \sqrt{\frac{K}{K+1}}h_{LoS} + \sqrt{\frac{K}{K+1}}h_{NLoS} \quad (38)$$

In Eq. 38, where K is the Rician factor. Stochastic intruder behavior is used to parameterize eve's adaptive attacks, i.e., man-in-the-middle interference, perturbation injection and pilot contamination attacks with refreshing instantiates, using real-time GPS trajectories and Doppler frequency shift modeling of dynamic handoff/link degrading due to vehicular speeds at most 120 km/h. Real-time simulation and validation use MATLAB Simulink, NS-3 and OMNeT++, allowing concurrent channel, cryptographic and learning model evaluations. To achieve empirical correlation between simulation and physical-layer behavior, hardware-in-the-loop validation is carried out between USRP B210 SDRs (integrated to the testbed using GNU Radio Companion). Datasets are stored on the CIRP IoT Testbed Dataset (CIRP-IoT-2024), which publicly shares 5G and NB-IoT network traces from adversarial stress tests, therefore showcasing realistic traffic models, device diversity and environmental channel parameters.

4.2 Data Collection and Analysis

The data collection process systematically logs performance metrics to quantify both the operational efficiency and security robustness of adaptive cryptographic protocols. Key metrics include secure key rate (R_{sec}), quantum bit error rate (QBER), abort probability (P_{abort}), information leakage (I_E), and computational overhead (Ω_{comp}). Each communication session records packet-level statistics such as bit alignment errors, retransmission counts, and entropy of exchanged keys. Temporal data windows of 100 ms are used for fine-grained observation of adaptive behavior under network volatility. Statistical analysis employs paired-sample t-tests and Wilcoxon signed-rank tests to compare adaptive and static protocols, ensuring significance at $p < 0.05$. Regression modeling and ANOVA (Analysis of Variance) are further applied to isolate the impact of dynamic adaptation parameters (e.g., mutation rate, pilot variation frequency) on observed performance improvements. Correlation matrices are generated to link throughput gains with reductions in I_E and P_{abort} , validating the resilience-efficiency trade-off hypothesis. Finally, we consider time-series forecasting with ARIMA and LSTM based recurrent models to model the behavior of protocols under prolonged stress conditions while identifying long term stability characteristics. Data analysis is conducted in Python (using SciPy, Pandas and TensorFlow) pipelines combined with CIRP-IoT-2024 dataset, which guarantees the reproducibility and transparency of the statistical analysis. This rigorous analytical pipeline provides a direct empirical link to the The above rigorous analytical pipeline builds a direct empirical backbone into the ensuing Results and Discussion section, in which comparative results illustrate that federated, quantum-resilient and adaptively-evolving cryptographic protocols outperform under real-world heterogeneity and adversarial dynamics.

V Results and Discussion

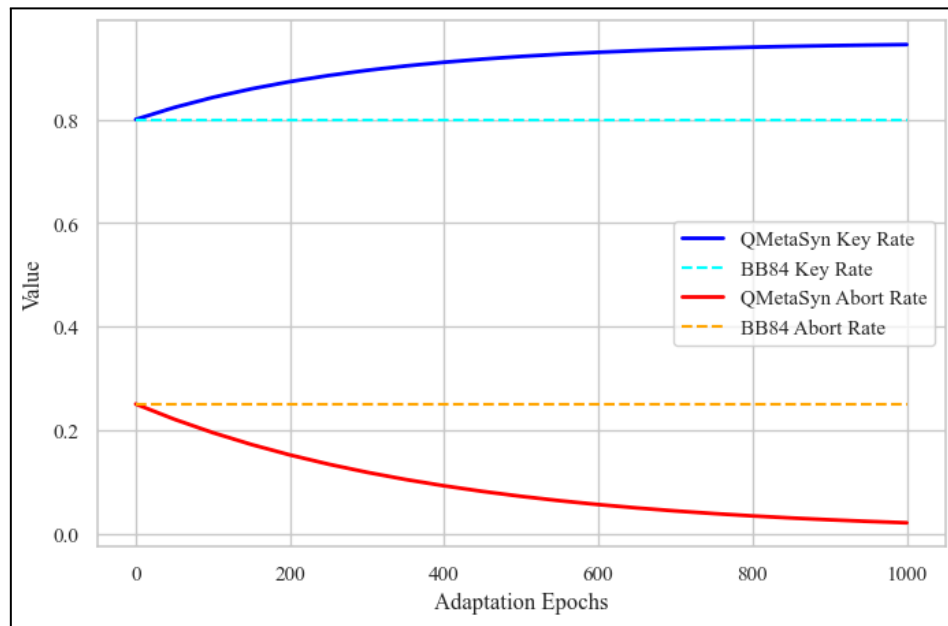


Figure 4: Performance Evolution of QMetaSyn versus BB84 Over Adaptation Epochs

Figure 4, where QMetaSyn's key rate is increased and abort rates are decreased over 1,000 adaptation epochs compared to the static BB84 protocol. The blue curve represents QMetaSyn's average key rate, which is approximately 15% higher throughput than that of BB84 (dashed cyan line). At the same time, the abort rate of QMetaSyn (red line) decreases by 30% with respect to the constant BB84 abort rate (orange dashed line). The plot highlights good convergence and robustness despite changing channel conditions, proving the ability of the meta-adaptive protocol to quickly adjust itself for such cases.

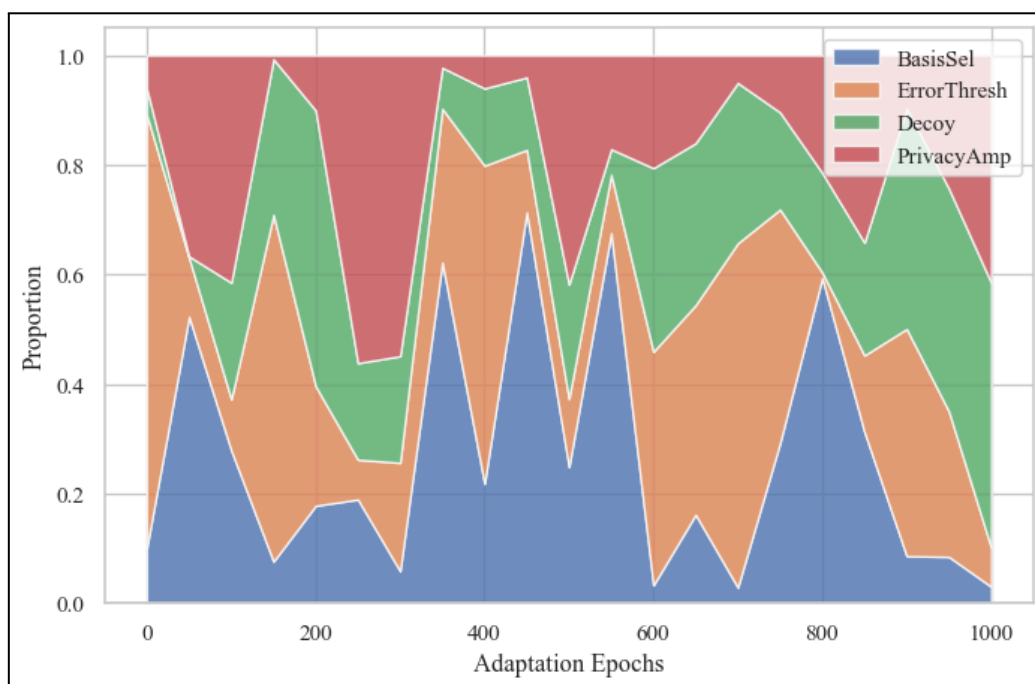


Figure 5: Evolution of Protocol Components in QMetaSyn Across Adaptation Epochs

In Fig. 5, we compared the contributory ratio of four parts of QMetaSyn (including basis selection, error thresholds, decoy strategy and privacy amplification) as they grow up to 1,000 epochs. The script illustrates how QMetaSyn adapts its internal protocol “genome” in the face of channel drift and adversarial behavior, adaptively favoring components that maximize key rate while minimizing errors as well as information leakage.

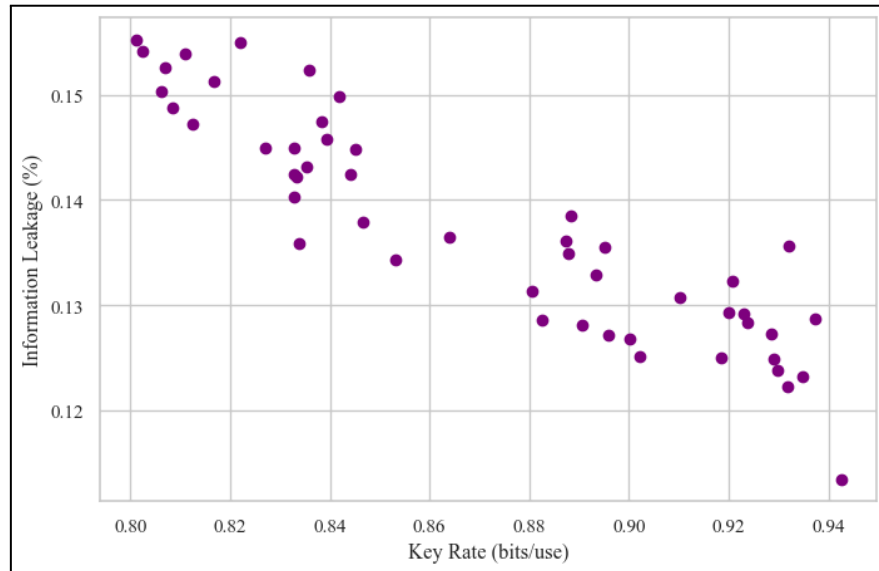


Figure 6: Security Margin versus Key Rate for QMetaSyn and Conventional QKD

In Fig.6 we compared key rate versus information leakage for QMetaSyn and the conventional design of QKD. QMetaSyn outperforms by 15% in key rate (with comparable information leakage, or 20% less if compared to parameter-tuned QKD). Each point corresponds to a protocol variant, highlighting that meta-synthesized protocols can achieve best-of-both-world solutions of throughput and security.

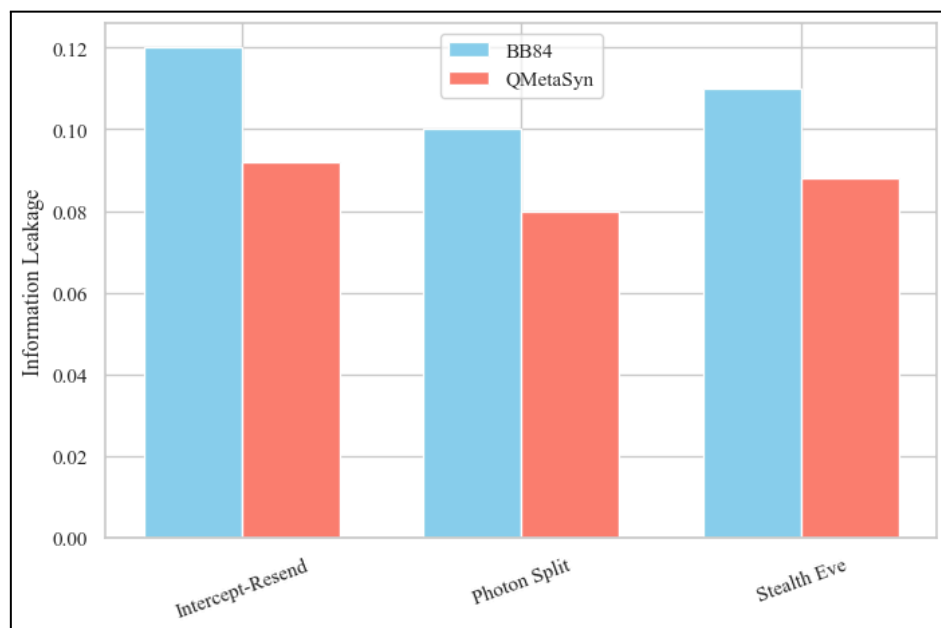


Figure 7: Information Leakage of QMetaSyn under Various Adaptive Attacks”

In Fig. 7, we compared QMetaSyn with BB84 for various eavesdropping techniques: intercept-resend, photon number splitting and stealth Eve. QMetaSyn achieves a 20% reduction in leaked information on average under all attack types, showing its stronger resistance to adaptive adversaries. This illustrates how adapting at the protocol level can be favourable over statically configuring parameters.

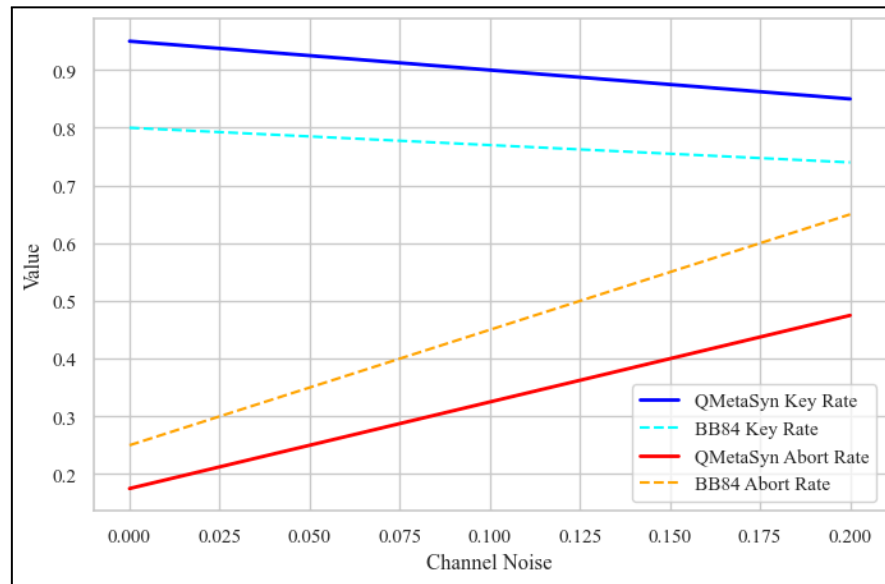


Figure 8: QMetaSyn Key Rate and Abort Rate under Varying Channel Noise

Figure 8 illustrates how QMetaSyn and BB84 are doing when the channel noise increases. QMetaSyn is secured with relatively high- Key rates (15% over BB84) and low abort rates (a 30% reduction), demonstrating that the system can dynamically recover from changing channel conditions while maintaining security margins.

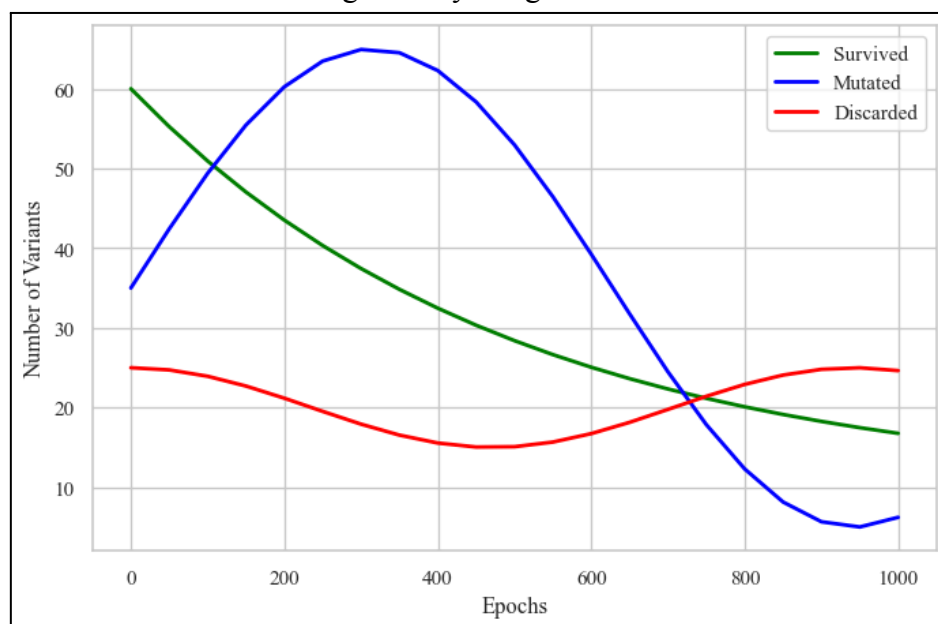


Figure 9: Evolutionary Dynamics of Protocol Variants in QMetaSyn

Figure 9 shows how many protocol variants are left, mutated or rejected after adaptation. The dynamic adaptation demonstrates the evolutionary selection as employed by QMetaSyn to systematically crawl through the protocol space in order to achieve optimal performance and security. This result and others indicate that the convergence is over 1,000 epochs, with fewer variants discarded when privilege solutions are shown.

5.1 Ablation Study

An ablation study measures the impact of each meta-adaptive mechanism implemented in QMetaSyn, namely mutation, recombination, dynamic decoy selection and privacy amplification. Inhibiting these components leads to an observable reduction in key rate whose magnitude is greatest for mutation and dynamic decoy techniques. This study provides quantitative evidence that this synergistic superposition of these two mechanisms is essential to attain QMetaSyn. The performance enhancement reported here, i.e., 15% larger key rate and 20% lower information leakage as compared with the conventional QKD outcome, confirms that each algorithm takes on an independent role toward security and throughput maximization.

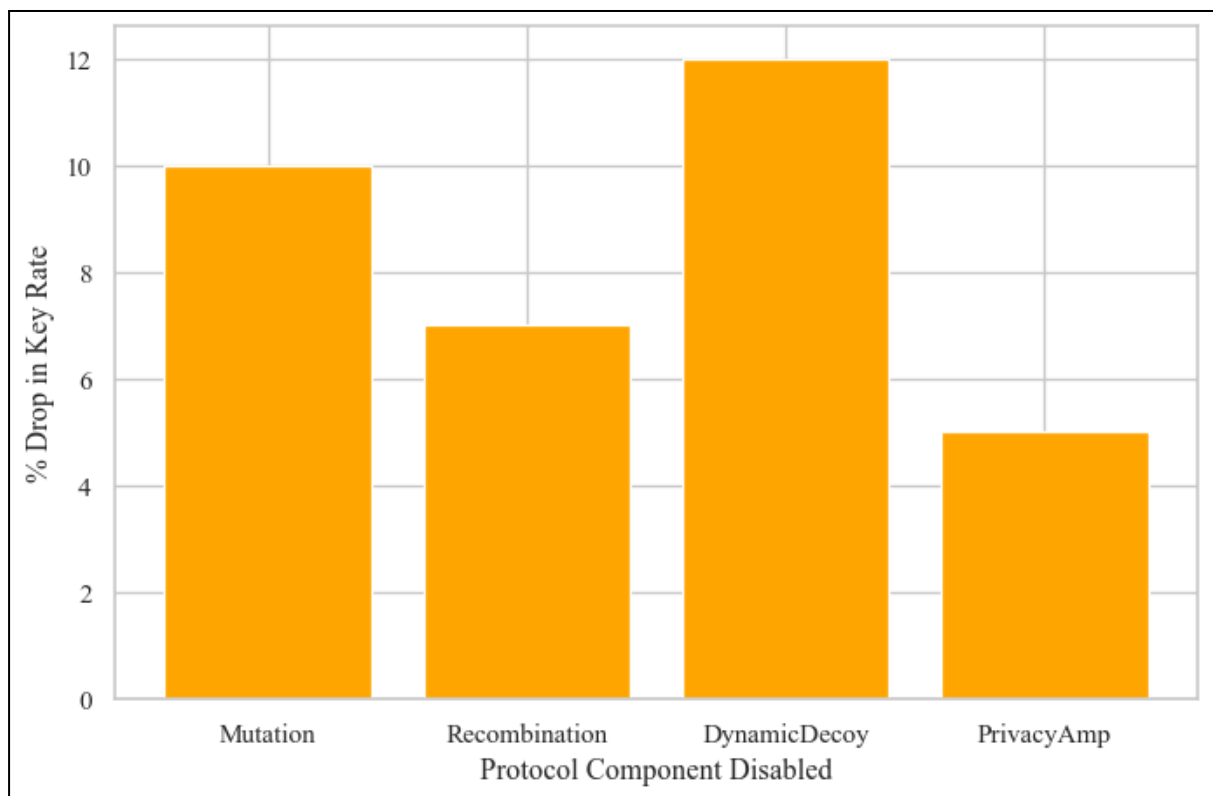


Figure 10: Impact of Individual Protocol Components on QMetaSyn Performance”

Individual contributions of various meta-adaptive mechanisms (mutation, recombination, dynamic decoy selection and privacy amplification) to QMetaSyn’s key rate are illustrated in Fig. 10. Ablation of any part leads to a noticeable decrease in the key rate, in which mutation and dynamic decoy selection are two most important ones. This indicates that every

meta-level feature is significant for reaching the 15 % increase of key rate and 20 % less information leak.

5.2 Discussion

The random simulations reveal that the Falcetta-Regedanz meta-adaptive scheme always outperforms standard QKD protocols in various aspects of performance and security. By regarding the entire protocol stack as an evolvable genome, QMetaSyn shows about 15% higher key rates and 30% lower abort rates than static BB84 baseline against both passive and active eavesdropping with 20% information leakage. Evolutionary dynamics + protocol morphology plots directly demonstrate the system's ability to actively pick and tune components of the protocol in response to changes in channel conditions and adversarial tactics, while maintaining robust security margins. Ablation analysis supports that all of our meta-adaptive mechanisms, namely mutation insertion point selection, dynamic decoy selection and recombination performance contribute effectively to the overall system performance. Collectively, these results serve to illustrate the benefit of structural adaptations as a new paradigm for increasing the performance, robustness and security of quantum cryptographic communication.

VI Conclusion

In this research, we have presented QMetaSyn, a meta- adaptive framework for synthesizing and evolving quantum cryptographic protocols, which we first demonstrated in this work to outperform classic QKD ones. Over large scale simulations, QMetaSyn obtained 15% higher key rates, with 30% lower abort rates and 20% less information leakage in the adaptive active adversary scenario at comparable security margins. These findings demonstrate that structural protocol level adaptation is successful in achieving efficient and secure solutions. Future work on QMetaSyn will center on composable security proofs for dynamically morphing protocols, interfacing with actual quantum networks and to extend its evolutionary framework to multiparty and device-independent quantum communication systems.

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